A Case Study Complete Formalization

This appendix reports the complete formalization of the case study presented in Section 5, derived from the Sepsis Cases event log in the XES format using the Inductive Miner by the PM4Py[4] library.

A.1 PDDL Domain File

This is the content of the `domain.pddl` file generated for the case study.

```
(define (domain sepsis)
  (:requirements :strips :typing
    :universal-preconditions
    :conditional-effects
    :negative-preconditions)

  (:types
    activity leucocytes_type
    lacticacid_type crp_type)

  (:constants
    admission_nc crp er_registration
    er_sepsis_triage er_triage
    iv_antibiotics iv_liquid lacticacid
    leucocytes release_a release_b
    return_er tau_1 tau_10 tau_11
    tau_12 tau_13 tau_14 tau_15 tau_16
    tau_17 tau_18 tau_19 tau_2 tau_20
    tau_21 tau_3 tau_4 tau_5 tau_6
    tau_7 tau_8 tau_9 - activity
    gte_0_2_lte_0_75 gte_0_75_lte_2_15
    gte_2_15 lte_0_2 - lacticacid_type
    gte_147_5 gte_28_5_lte_40_5
    gte_40_5_lte_57_5 gte_57_5_lte_147_5
    lte_28_5 - crp_type
    gte_12_15_lte_13_95 gte_13_95
    gte_3_4_lte_7_15 gte_7_15_lte_12_15
    lte_3_4 - leucocytes_type)

  (:predicates
    (completed ?a - activity)
    (enabled ?a - activity)
    (lacticacid ?v - lacticacid_type)
    (diagnosticlacticacid)
    (crp ?v - crp_type)
    (leucocytes ?v - leucocytes_type))

  (:action exec_admission_nc
   :parameters ()
   :precondition (and
      (enabled admission_nc)
      (completed tau_7))
   :effect (and
      (completed admission_nc)
      (enabled tau_15)
      (enabled tau_7)
      (not (enabled admission_nc))))

  (:action exec_crp-gte_147_5
   :parameters ()
   :precondition (enabled crp)
   :effect (and
      (completed crp)
      (forall (?old - crp_type)
       (when (crp ?old)
         (not (crp ?old))))
      (crp gte_147_5)
      (enabled tau_14)
      (enabled tau_1)
      (not (enabled crp))))

  (:action exec_crp-gte_28_5_lte_40_5
   :parameters ()
   :precondition (enabled crp)
   :effect (and
      (completed crp)
      (forall (?old - crp_type)
       (when (crp ?old)
         (not (crp ?old))))
      (crp gte_28_5_lte_40_5)
      (enabled tau_14)
      (enabled tau_1)
      (not (enabled crp))))

  (:action exec_crp-gte_40_5_lte_57_5
   :parameters ()
   :precondition (enabled crp)
   :effect (and
      (completed crp)
      (forall (?old - crp_type)
       (when (crp ?old)
         (not (crp ?old))))
      (crp gte_40_5_lte_57_5)
      (enabled tau_14)
      (enabled tau_1)
      (not (enabled crp))))

  (:action exec_crp-gte_57_5_lte_147_5
   :parameters ()
   :precondition (enabled crp)
   :effect (and
      (completed crp)
      (forall (?old - crp_type)
       (when (crp ?old)
         (not (crp ?old))))
      (crp gte_57_5_lte_147_5)
      (enabled tau_14)
      (enabled tau_1)
      (not (enabled crp))))

  (:action exec_crp-lte_28_5
   :parameters ()
   :precondition (enabled crp)
   :effect (and
      (completed crp)
      (forall (?old - crp_type)
```

[4] https://processintelligence.solutions/static/api/2.7.17/index.html

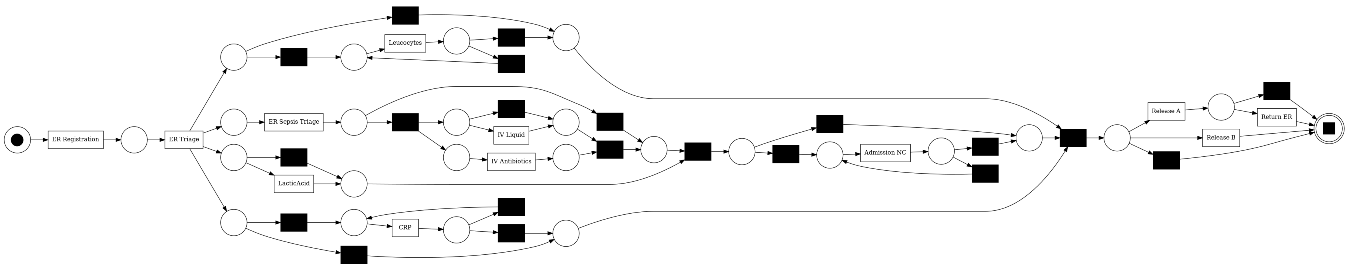

Figure 3: Petri net discovered by the Inductive Miner from the Sepsis Cases event log. Silent transitions are reported in black.

```
      (when (crp ?old)
        (not (crp ?old))))
     (crp lte_28_5)
     (enabled tau_14)
     (enabled tau_1)
     (not (enabled crp))))

(:action exec_er_registration
 :parameters ()
 :precondition (enabled er_registration)
 :effect (and
     (completed er_registration)
     (enabled er_triage)
     (not (enabled er_registration))))

(:action exec_er_sepsis_triage_v1
 :parameters ()
 :precondition (and
     (enabled er_sepsis_triage)
     (crp gte_57_5_lte_147_5))
 :effect (and
     (completed er_sepsis_triage)
     (enabled tau_9)
     (enabled tau_2)
     (not (enabled er_sepsis_triage))))

(:action exec_er_sepsis_triage_v2
 :parameters ()
 :precondition (and
     (enabled er_sepsis_triage)
     (crp gte_147_5))
 :effect (and
     (completed er_sepsis_triage)
     (enabled tau_9)
     (enabled tau_2)
     (not (enabled er_sepsis_triage))))

(:action exec_er_triage
 :parameters ()
 :precondition (enabled er_triage)
 :effect (and
     (completed er_triage)
     (enabled tau_6)
     (enabled tau_5)
     (enabled tau_18)
     (enabled tau_21)
```

```
     (enabled lacticacid)
     (enabled tau_4)
     (enabled er_sepsis_triage)
     (not (enabled er_triage))))

(:action exec_iv_antibiotics
 :parameters ()
 :precondition (and
     (completed tau_9)
     (enabled iv_antibiotics))
 :effect (and
     (completed iv_antibiotics)
     (enabled tau_16)
     (not (enabled iv_antibiotics))))

(:action exec_iv_liquid
 :parameters ()
 :precondition (and
     (enabled iv_liquid)
     (diagnosticlacticacid))
 :effect (and
     (completed iv_liquid)
     (enabled tau_16)
     (not (enabled iv_liquid))))

(:action exec_lacticacid-gte_0_2_lte_0_75
 :parameters ()
 :precondition (and
     (completed er_triage)
     (enabled lacticacid))
 :effect (and
     (completed lacticacid)
     (forall (?old - lacticacid_type)
      (when (lacticacid ?old)
        (not (lacticacid ?old))))
     (lacticacid gte_0_2_lte_0_75)
     (enabled tau_17)
     (not (enabled lacticacid))))

(:action exec_lacticacid-gte_0_75_lte_2_15
 :parameters ()
 :precondition (and
     (completed er_triage)
     (enabled lacticacid)
 )
 :effect (and
```

```
    (completed lacticacid)
    (forall (?old - lacticacid_type)
     (when (lacticacid ?old)
       (not (lacticacid ?old))))
    (lacticacid gte_0_75_lte_2_15)
    (enabled tau_17)
    (not (enabled lacticacid))))

(:action exec_lacticacid-gte_2_15
 :parameters ()
 :precondition (and
    (completed er_triage)
    (enabled lacticacid))
 :effect (and
    (completed lacticacid)
    (forall (?old - lacticacid_type)
     (when (lacticacid ?old)
       (not (lacticacid ?old))))
    (lacticacid gte_2_15)
    (enabled tau_17)
    (not (enabled lacticacid))))

(:action exec_lacticacid-lte_0_2
 :parameters ()
 :precondition (and
    (completed er_triage)
    (enabled lacticacid))
 :effect (and
    (completed lacticacid)
    (forall (?old - lacticacid_type)
     (when (lacticacid ?old)
       (not (lacticacid ?old))))
    (lacticacid lte_0_2)
    (enabled tau_17)
    (not (enabled lacticacid))))

(:action exec_leucocytes-gte_12_15_lte_13_95
 :parameters ()
 :precondition (enabled leucocytes)
 :effect (and
    (completed leucocytes)
    (forall (?old - leucocytes_type)
     (when (leucocytes ?old)
       (not (leucocytes ?old))))
    (leucocytes gte_12_15_lte_13_95)
    (enabled tau_8)
    (enabled tau_20)
    (not (enabled leucocytes))))

(:action exec_leucocytes-gte_13_95
 :parameters ()
 :precondition (enabled leucocytes)
 :effect (and
    (completed leucocytes)
    (forall (?old - leucocytes_type)
     (when (leucocytes ?old)
       (not (leucocytes ?old))))
    (leucocytes gte_13_95)
    (enabled tau_8)
    (enabled tau_20)
    (not (enabled leucocytes))))

(:action exec_leucocytes-gte_3_4_lte_7_15
 :parameters ()
 :precondition (enabled leucocytes)
 :effect (and
    (completed leucocytes)
    (forall (?old - leucocytes_type)
     (when (leucocytes ?old)
       (not (leucocytes ?old))))
    (leucocytes gte_3_4_lte_7_15)
    (enabled tau_8)
    (enabled tau_20)
    (not (enabled leucocytes))))

(:action exec_leucocytes-gte_7_15_lte_12_15
 :parameters ()
 :precondition (enabled leucocytes)
 :effect (and
    (completed leucocytes)
    (forall (?old - leucocytes_type)
     (when (leucocytes ?old)
       (not (leucocytes ?old))))
    (leucocytes gte_7_15_lte_12_15)
    (enabled tau_8)
    (enabled tau_20)
    (not (enabled leucocytes))))

(:action exec_leucocytes-lte_3_4
 :parameters ()
 :precondition (enabled leucocytes)
 :effect (and
    (completed leucocytes)
    (forall (?old - leucocytes_type)
     (when (leucocytes ?old)
       (not (leucocytes ?old))))
    (leucocytes lte_3_4)
    (enabled tau_8)
    (enabled tau_20)
    (not (enabled leucocytes))))

(:action exec_release_a
 :parameters ()
 :precondition (and
    (enabled release_a)
    (completed tau_19))
 :effect (and
    (completed release_a)
    (enabled return_er)
    (enabled tau_3)
    (not (enabled release_a))))

(:action exec_release_b_v1
 :parameters ()
 :precondition (and
    (crp gte_147_5)
    (enabled release_b)
    (completed tau_19)
```

```
    (completed iv_antibiotics))
 :effect (and
    (completed release_b)
    (not (enabled release_b))))

(:action exec_return_er_v1
 :parameters ()
 :precondition (and
    (leucocytes gte_3_4_lte_7_15)
    (enabled return_er)
    (completed leucocytes)
    (completed release_a))
 :effect (and
    (completed return_er)
    (not (enabled return_er))))

(:action exec_tau_1
 :parameters ()
 :precondition (and
    (completed crp)
    (enabled tau_1))
 :effect (and
    (completed tau_1)
    (enabled crp)
    (not (enabled tau_1))))

(:action exec_tau_10
 :parameters ()
 :precondition (and
    (completed tau_17)
    (enabled tau_10))
 :effect (and
    (completed tau_10)
    (enabled admission_nc)
    (not (enabled tau_10))))

(:action exec_tau_11
 :parameters ()
 :precondition (and
    (completed tau_19)
    (enabled tau_11))
 :effect (and
    (completed tau_11)
    (not (enabled tau_11))))

(:action exec_tau_12
 :parameters ()
 :precondition (and
    (completed tau_9)
    (enabled tau_12))
 :effect (and
    (completed tau_12)
    (enabled tau_16)
    (not (enabled tau_12))))

(:action exec_tau_13
 :parameters ()
 :precondition (and
    (completed tau_17)
    (enabled tau_13))
 :effect (and
    (completed tau_13)
    (enabled tau_19)
    (not (enabled tau_13))))

(:action exec_tau_14
 :parameters ()
 :precondition (and
    (completed crp)
    (enabled tau_14))
 :effect (and
    (completed tau_14)
    (enabled tau_19)
    (not (enabled tau_14))))

(:action exec_tau_15
 :parameters ()
 :precondition (and
    (enabled tau_15)
    (completed admission_nc))
 :effect (and
    (completed tau_15)
    (enabled tau_19)
    (not (enabled tau_15))))

(:action exec_tau_16
 :parameters ()
 :precondition (and
    (enabled tau_16)
    (completed iv_antibiotics))
 :effect (and
    (completed tau_16)
    (enabled tau_17)
    (not (enabled tau_16))))

(:action exec_tau_17
 :parameters ()
 :precondition (and
    (completed tau_4)
    (enabled tau_17))
 :effect (and
    (completed tau_17)
    (enabled tau_13)
    (enabled tau_10)
    (not (enabled tau_17))))

(:action exec_tau_18
 :parameters ()
 :precondition (and
    (enabled tau_18)
    (completed er_triage))
 :effect (and
    (completed tau_18)
    (enabled crp)
    (not (enabled tau_18))))

(:action exec_tau_19
 :parameters ()
```

```
 :precondition (and
    (completed tau_21)
    (enabled tau_19))
 :effect (and
    (completed tau_19)
    (enabled release_a)
    (enabled release_b)
    (enabled tau_11)
    (not (enabled tau_19))))

(:action exec_tau_2
 :parameters ()
 :precondition (and
    (enabled tau_2)
    (completed er_sepsis_triage))
 :effect (and
    (completed tau_2)
    (enabled tau_17)
    (not (enabled tau_2))))

(:action exec_tau_20
 :parameters ()
 :precondition (and
    (completed leucocytes)
    (enabled tau_20))
 :effect (and
    (completed tau_20)
    (enabled tau_19)
    (not (enabled tau_20))))

(:action exec_tau_21
 :parameters ()
 :precondition (and
    (completed er_triage)
    (enabled tau_21))
 :effect (and
    (completed tau_21)
    (enabled tau_19)
    (not (enabled tau_21))))

(:action exec_tau_3
 :parameters ()
 :precondition (and
    (enabled tau_3)
    (completed release_a))
 :effect (and
    (completed tau_3)
    (not (enabled tau_3))))

(:action exec_tau_4
 :parameters ()
 :precondition (and
    (completed er_triage)
    (enabled tau_4))
 :effect (and
    (completed tau_4)
    (enabled tau_17)
    (not (enabled tau_4))))

(:action exec_tau_5
 :parameters ()
 :precondition (and
    (enabled tau_5)
    (completed er_triage))
 :effect (and
    (completed tau_5)
    (enabled tau_19)
    (not (enabled tau_5))))

(:action exec_tau_6
 :parameters ()
 :precondition (and
    (enabled tau_6)
    (completed er_triage))
 :effect (and
    (completed tau_6)
    (enabled leucocytes)
    (not (enabled tau_6))))

(:action exec_tau_7
 :parameters ()
 :precondition (and
    (enabled tau_7)
    (completed admission_nc))
 :effect (and
    (completed tau_7)
    (enabled admission_nc)
    (not (enabled tau_7))))

(:action exec_tau_8
 :parameters ()
 :precondition (and
    (enabled tau_8)
    (completed leucocytes))
 :effect (and
    (completed tau_8)
    (enabled leucocytes)
    (not (enabled tau_8))))

(:action exec_tau_9
 :parameters ()
 :precondition (and
    (completed er_sepsis_triage)
    (enabled tau_9))
 :effect (and
    (completed tau_9)
    (enabled tau_12)
    (enabled iv_liquid)
    (enabled iv_antibiotics)
    (not (enabled tau_9))))
)
```

A.2 PDDL Problem File

This is the content of the `problem.pddl` file generated for the case study.

```
(define (problem sepsis_prediction)
  (:domain sepsis)
```

```
  (:init
    (completed er_registration)
    (enabled er_triage)
  )

  (:goal
    (and
      (completed er_sepsis_triage)
      (crp gte_147_5))
  )
)
```

B Case Study Code

The event logs presented in the paper, source code, and instructions for using the methodology and replicating the experiments are available in the zipped folder provided as supplementary material.

If the paper is accepted, the material will be uploaded to GitHub and Zenodo.

C Complete Evaluation Results

We also report in the following pages the complete evaluation results for the *situation* and *suffix prediction* tasks, which are not included in the paper due to page limits.

Table 7: Complete results for the situation prediction for all search strategies. The discovery algorithm is fixed to *Inductive Miner* for all experiments.

Event Log	**Search Strategy**	**Samples**	**Solvability (%)**	**Structural Unsolvability (%)**	**Timeouts (%)**	**Avg. Length Difference**	**Search Time (s)**	**Grounding Time (s)**	**Expanded Nodes**	**Memory (MB)**	**Solution Length**
BPI12 A	*A* (Blind)*	500	100.00	0.00	0.00	-0.67	0.0003	0.063	8	10.06	3.50
BPI12 A	*A** (h_{ff})	500	100.00	0.00	0.00	-0.67	0.0004	0.064	5	10.06	3.50
BPI12 A	*A* (LM-Cut)*	500	100.00	0.00	0.00	-0.67	0.0004	0.063	5	10.06	3.50
BPI12 A	*Greedy* (h_{ff})	500	100.00	0.00	0.00	-0.67	0.0003	0.064	5	10.06	3.50
BPI12 A	*LAMA 2011*	500	100.00	0.00	0.00	-0.67	0.0036	0.064	4	10.19	3.50
BPI12 O	*A* (Blind)*	500	73.20	26.80	0.00	-1.20	0.0003	0.028	15	10.06	4.70
BPI12 O	*A** (h_{ff})	500	73.20	26.80	0.00	-1.20	0.0004	0.029	4	10.06	4.70
BPI12 O	*A* (LM-Cut)*	500	73.20	26.80	0.00	-1.20	0.0004	0.028	4	10.06	4.70
BPI12 O	*Greedy* (h_{ff})	500	73.20	26.80	0.00	-1.20	0.0003	0.028	4	10.06	4.70
BPI12 O	*LAMA 2011*	500	73.20	26.80	0.00	-0.84	0.0038	0.028	4	10.19	5.20
BPI12 W	*A* (Blind)*	500	31.00	69.00	0.00	-1.27	0.0002	0.304	6	10.06	4.50
BPI12 W	*A** (h_{ff})	500	31.00	69.00	0.00	-1.27	0.0002	0.303	2	10.06	4.50
BPI12 W	*A* (LM-Cut)*	500	31.00	69.00	0.00	-1.27	0.0002	0.302	2	10.06	4.50
BPI12 W	*Greedy* (h_{ff})	500	31.00	69.00	0.00	-1.27	0.0002	0.303	2	10.06	4.50
BPI12 W	*LAMA 2011*	500	31.00	69.00	0.00	-1.27	0.0035	0.300	1	10.19	4.50
BPI12 WC	*A* (Blind)*	500	80.60	19.40	0.00	-1.94	0.0003	0.047	10	10.06	4.50
BPI12 WC	*A** (h_{ff})	500	80.60	19.40	0.00	-1.94	0.0004	0.048	4	10.06	4.50
BPI12 WC	*A* (LM-Cut)*	500	80.60	19.40	0.00	-1.94	0.0004	0.047	4	10.06	4.50
BPI12 WC	*Greedy* (h_{ff})	500	80.60	19.40	0.00	-1.94	0.0004	0.048	4	10.06	4.50
BPI12 WC	*LAMA 2011*	500	80.60	19.40	0.00	-1.94	0.0042	0.048	4	10.19	4.50
BPI13 CP	*A* (Blind)*	500	100.00	0.00	0.00	-1.53	0.0002	0.013	2	10.06	1.00
BPI13 CP	*A** (h_{ff})	500	100.00	0.00	0.00	-1.53	0.0002	0.013	2	10.06	1.00
BPI13 CP	*A* (LM-Cut)*	500	100.00	0.00	0.00	-1.53	0.0002	0.013	2	10.06	1.00
BPI13 CP	*Greedy* (h_{ff})	500	100.00	0.00	0.00	-1.53	0.0002	0.013	2	10.06	1.00
BPI13 CP	*LAMA 2011*	500	100.00	0.00	0.00	-1.53	0.0029	0.013	1	10.19	1.00
BPI13 I	*A* (Blind)*	500	48.80	51.20	0.00	-0.69	0.0002	0.018	6	10.06	3.70
BPI13 I	*A** (h_{ff})	500	48.80	51.20	0.00	-0.69	0.0003	0.018	2	10.06	3.70
BPI13 I	*A* (LM-Cut)*	500	48.80	51.20	0.00	-0.69	0.0003	0.018	2	10.06	3.70
BPI13 I	*Greedy* (h_{ff})	500	48.80	51.20	0.00	-0.69	0.0003	0.018	2	10.06	3.70
BPI13 I	*LAMA 2011*	500	48.80	51.20	0.00	-0.37	0.0032	0.018	2	10.19	4.30
Sepsis	*A* (Blind)*	500	100.00	0.00	0.00	-1.60	0.0056	0.030	893	10.20	4.60
Sepsis	*A** (h_{ff})	500	100.00	0.00	0.00	-1.44	0.0006	0.030	7	10.12	5.70
Sepsis	*A* (LM-Cut)*	500	100.00	0.00	0.00	-1.60	0.0008	0.029	6	10.08	4.60
Sepsis	*Greedy* (h_{ff})	500	100.00	0.00	0.00	-1.60	0.0006	0.029	6	10.12	4.60
Sepsis	*LAMA 2011*	500	100.00	0.00	0.00	-1.60	0.0455	0.029	5	10.25	4.80

Table 8: Complete suffix prediction results across all discovery algorithms. The search strategy is fixed to *A** search with a *blind* heuristic. The table reports *solvability* (S), *structural unsolvability* (U_s), *timeouts* (U_t), *exact matches* (E), *normalized DL similarity* (Sim), and *average DL distance* ($Dist$). Planner's timeout was set to *30 s*.

Event Log	Discovery	Samples	S (%)	U_s (%)	U_t (%)	E (%)	Sim (%)	$Dist$
BPI12 A	*Alpha*	500	100.0	0.0	0.0	71.4	94.3	14.0
	Heuristics	500	100.0	0.0	0.0	71.0	94.3	14.1
	ILP	500	100.0	0.0	0.0	45.4	97.6	10.2
	Inductive	500	100.0	0.0	0.0	71.4	95.2	7.9
BPI12 O	*Alpha*	500	100.0	0.0	0.0	74.0	89.3	13.8
	Heuristics	500	100.0	0.0	0.0	72.8	88.0	14.0
	ILP	500	100.0	0.0	0.0	27.8	86.2	23.7
	Inductive	500	100.0	0.0	0.0	31.8	86.3	21.1
BPI12 W	*Alpha*	500	100.0	0.0	0.0	2.2	25.8	81.1
	Heuristics	500	100.0	0.0	0.0	18.4	27.7	68.2
	ILP	500	100.0	0.0	0.0	9.6	27.4	71.2
	Inductive	500	91.4	8.6	0.0	2.4	26.0	75.4
BPI12 WC	*Alpha*	500	94.2	5.8	0.0	6.2	13.2	28.0
	Heuristics	500	100.0	0.0	0.0	9.4	13.9	31.2
	ILP	500	100.0	0.0	0.0	9.4	13.9	31.2
	Inductive	500	100.0	0.0	0.0	9.4	13.9	31.2
BPI13 CP	*Alpha*	500	62.4	37.6	0.0	0.0	87.1	21.3
	Heuristics	500	100.0	0.0	0.0	41.8	90.9	21.6
	ILP	500	100.0	0.0	0.0	41.8	90.4	24.0
	Inductive	500	62.2	37.8	0.0	56.6	90.5	12.4
BPI13 I	*Alpha*	500	76.0	24.0	0.0	22.1	95.1	25.0
	Heuristics	500	100.0	0.0	0.0	48.8	97.1	28.0
	ILP	500	96.8	3.2	0.0	15.5	93.6	47.7
	Inductive	500	46.4	53.6	0.0	52.6	95.1	18.1
Sepsis	*Alpha*	326	100.0	0.0	0.0	23.9	96.5	31.2
	Heuristics	326	83.7	0.0	16.3	34.4	97.2	18.4
	ILP	326	100.0	0.0	0.0	15.0	96.3	28.9
	Inductive	326	87.7	0.0	12.3	15.7	95.9	30.2